National University of Computer and Emerging Sciences
Lahore Campus

Engineering Economics (MG3008)

Date: May 27th 2024

Course Instructor(s)

Mr. Syed Haris Mujtaba

Final Exam

Total Time (Hrs): 3
Total Marks: 115
Total Questions: 5

______________ Roll No   _____________ Section   ___________________________ Student Signature

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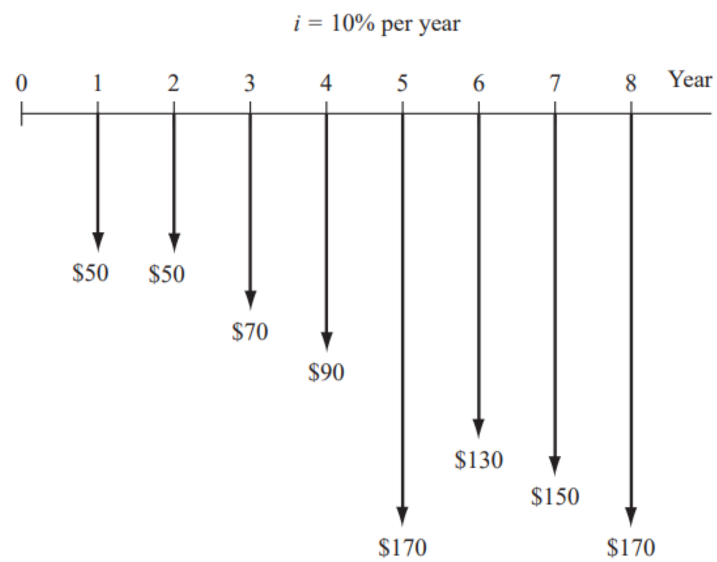
Attempt all the questions.
Solve everything on the answer sheet and show each step.
In those question where the table is provided fill the values in them and return the question paper with answer sheet.

CLO #: 1 Recognize the fundamental economic and financial considerations involved in engineering projects

Q1: **Find** the present worth in year 0 for the cash flows given below. Let i = 10% per year. [20 marks]



CLO #: 2 Interpret the concept of time value of money

Q2: For the cash flows given below, By applying the concept of time value of money, **determine** the equivalent uniform worth in years 1 through 5 at an interest rate of 18% per year, compounded monthly. [20 marks]

Year	1	2	3	4	5
Cash Flow, $	200,000	0	350,000	0	400,000

CLO #: 3 Analyze cost effectiveness of alternatives using engineering economy methods

Q3: The city is considering variable proposals regarding the disposal of used tires. An incremental B/C analysis was initiated but never completed. **Calculate** and fill in the missing blanks in the table and relate which alternative should be selected. [20 marks]

Alternative	PW of cost,$	PW of Benefits, $	PW of Disbenefits, $	B/C ratio	Incremental B/C when compared with Alternative			
					J	K	L	M
J	20	?	1	1.05	____	?	?	?
K	23	28	?	1.13		____	?	?
L	28	35	3	?			____	?
M	?	51	4	1.34				____

CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation

Q4-a: **Generate** a loan repayment table for an asset I= $950,000, interest is 12% per year for 5 of years; when equal end of year payments are made. [10marks]

N	Big-Bal	Interest	Payment	Principal	Remaining	Acc- Principal
0						
1						
2						
3						
4						
5						

Q4-b: The Marginal tax rates are given below, **enumerate** all the remaining columns of the table: [5 Marks]

Range	Tax Rate	Limit-From	Limit-Up-to	Maximum Tax of Range	Maximum Tax Incurred
150,000	8%				
200,000	12%				
200,000	18%				
150,000	24%				
2500,000	30%				
Above	40%				

CLO #: 4 Develop cash flow sequences that include the effects of taxes and depreciation

Q5-a: **Develop** a MACRS table with 150% DB also calculate the depreciation amount and book value for each year of an asset, bought for $900,000 and has a life of 6 years. [20 marks]

N	Bal	Dep1	Remain	N^*	Straight	Dep	Currency	BV
1								
2								
3								
4								
5								
6								
7								

Q5-b: Carry Forward from Q4 and Q5(a): From information calculated in table for interest in Q4-a; also use the MACRS table of Q5-a, and make the rest of the income statement using the tax table in Q4-b, with Revenue, Material, Labor and Overhead and their relative inflations given below: **Develop** the complete table of income statement. [20marks]

	"f"	0	1	2	3	4	5
Revenue	18	1,300,000					
Material	17	350,000					
Labor	16	280,000					
Overhead	16	150,000					
RMC							
Dep							
Ebit							
Interest							
EBT							
Tax							
EAT							

Formulas

$\left(\frac{F}{P}, i, n\right) => F = P(1+i)^n$	$\left(\frac{P}{F}, i, n\right) => P = \frac{F}{(1+i)^n}$
$\left(\frac{P}{A}, i, n\right) => P = A\frac{(1+i)^n - 1}{i(1+i)^n}$	$\left(\frac{A}{P}, i, n\right) => A = P\frac{i(1+i)^n}{(1+i)^n - 1}$
$\left(\frac{F}{A}, i, n\right) => F = A\frac{(1+i)^n - 1}{i}$	$\left(\frac{A}{F}, i, n\right) => A = F\frac{i}{(1+i)^n - 1}$
Arithmetic Gradient	$P = \frac{G}{i}\left[\frac{(1+i)^n - 1}{i(1+i)^n} - \frac{n}{(1+i)^n}\right]$
Arithmetic Gradient	$A = G\left[\frac{1}{i} - \frac{n}{(1+i)^n - 1}\right]$
Geometric Gradient $P_g = A_1\left\{\frac{1-\left(\frac{1+g}{1+i}\right)^n}{i-g}\right\} \quad g \neq i$ $P_g = A_1\left(\frac{n}{1+i}\right) \quad g = i$	
$Effective i = (1 + {}^{r}/_{m})^m - 1$	
Inflation Adjusted Interest Rate $i_f = i + f + if$	
$AW_k = -P(A/P, i, k) + S_k(A/F, i, k) - \left[\sum_{j=1}^{j=k} AOC_j(P/F, i, j)\right](A/P, i, k)$	